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ABSTRACT

Abstract.

1. INTRODUCTION

2. METHODS

2.1. *Accuracy of astroNN*

For 19784 stars, or $\sim 3.5\%$ of the sample, the astroNN [Fe/H] abundance estimate is $\gtrsim 0.5$ dex lower than the value provided by ASPCAP. The vast majority of these stars are on the lower-main sequence according to *Gaia* photometry, so we exclude stars with $\log(g) > 4$ in ASPCAP. In addition, 197 or $\sim 3\%$ of stars with APOKASC-2 asteroseismic ages are reported to be > 5 Gyr younger by astroNN. The APOKASC-2 age is unphysically high for many of these stars, often exceeding 15 Gyr. We identify prior mass loss as the cause of this discrepancy, so the astroNN ages are likely closer to the truth.

2.2. *Measurements of the Type Ia Supernova Delay Time Distribution*

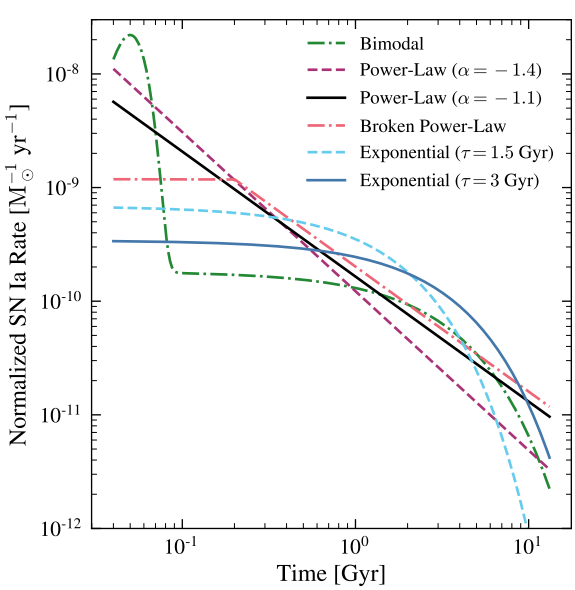

Figure 1. Delay time distribution functions

Table 1. Sources of the various forms of delay time distribution.

Name	Form	Source
Field Galaxy Power-Law	$t^{-1.1}$	Maoz & Graur (2017)
Galaxy Cluster Power-Law	$t^{-1.4}$	Maoz & Graur (2017)
Short Exponential	$e^{-t/1.5\,\mathrm{Gyr}}$	Schönrich & Binney (2009)

Schönrich & Binney (2009) take their DTD to be an exponential with a 1.5 Gyr timescale, though this isn't strictly observationally motivated.

Maoz & Graur (2017) compare volumetric SN Ia rates to the cosmic star formation history. They get a power law with a slope of -1.13 ± 0.06 for field galaxies, and a slope of $-1.39^{+0.32}_{-0.05}$ for galaxy clusters.

Heringer et al. (2019) study the distribution of SNe Ia in field galaxies and get a power-law index of $-1.34^{+0.19}_{-0.17}$ which is closer to Maoz's cluster DTD.

Strolger et al. (2020) perform maximum likelihood estimations based on the cosmic star formation history and the star formation histories of individual galaxies. In both cases their best fit functions were approximately exponential, though they started with a skew-normal function so could not have ended up with a power law even if they wanted to. They don't actually report the timescale of this exponential, but based on the plots I'd say it's about 1.5 - 2 (1.7?) Gyr. Their best fit parameters make no sense to me as they don't report their units.

Wiseman et al. (2021) measure the SN Ia rate in the redshift range $0.2 < z < 0.6$ and get a similar power law to Maoz & Graur (2017) with a slope of -1.13 ± 0.05. They also find that slower-declining SNe have a steeper DTD slope.

3. RESULTS

3.1. *One-Zone Models*

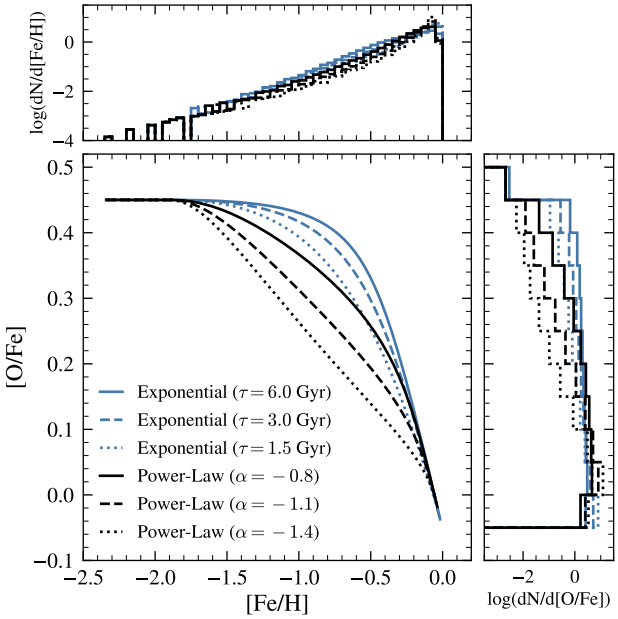

Figure 2. *Center:* Abundance tracks in the [Fe/H]-[O/Fe] plane for one-zone models with various DTD parameters. Tracks from models which assume a power-law DTD are in black, with dotted, dashed, and solid curves representing power-law slopes (α) of -0.8, -1.1, and -1.4, respectively. Tracks from models which assume an exponential DTD are in blue, with dotted, dashed, and solid curves representing exponential decay timescales (τ) of 1.5 Gyr, 3 Gyr, and 6 Gyr, respectively. In all cases a minimum delay time of 40 Myr is assumed. *Top:* MDFs generated from by one-zone models. *Right:* Distribution of [O/Fe] generated by the one-zone models.

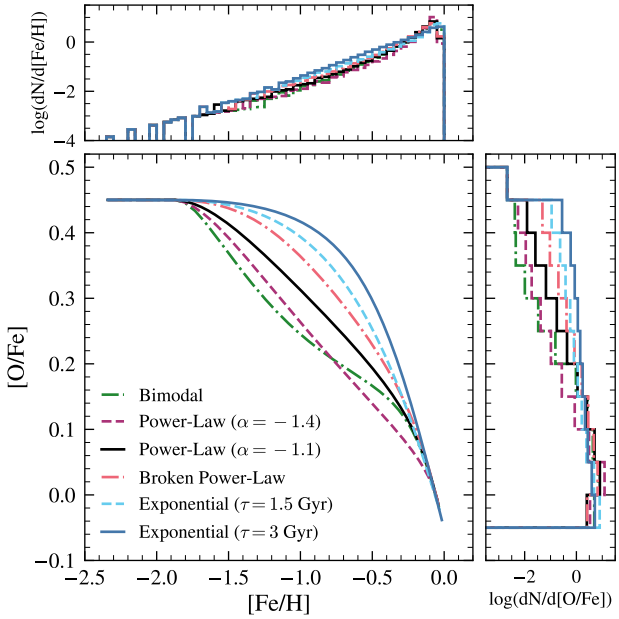

Figure 3. Similar to Figure 2 but with a different selection of DTDs. In all cases a minimum delay time of 40 Myr is assumed.

3.1.1. *The SN Ia Delay-Time Distribution*

Adjusting the parameters of the DTD affects how steeply the abundance track in [Fe/H]-[O/Fe] space falls from its initial "plateau" and the location of the "knee." Figure 2 shows abundance tracks from one-zone models which assume different slopes of a power-law DTD or different timescales of an exponential DTD. A steeper power-law slope results in a steeper decline in [O/Fe] and a lower knee, as does a shorter exponential timescale. The consequence is that a very shallow power-law produces similar observable signatures to a very steep exponential, which suggests these models may be difficult to distinguish in this parameter space.

Figure 3 compares the one-zone model outputs from the full range of DTDs we investigate in this paper. In general, the choice of DTD has the greatest effect on the location of the knee in [Fe/H]-[O/Fe] space (center panel) and the high-α end of the distribution of [O/Fe] (right panel). The DTD also has a smaller effect on the MDF in the range $-15.5 \lesssim$ [Fe/H] $\lesssim -0.5$ (top panel).

3.1.2. *The Minimum SN Ia Delay Time*

Figure 4 shows that the minimum SN Ia delay time has a much stronger effect on the abundance track and MDFs of models which assume a power-law DTD than models which assume an exponential, due to the much higher SN Ia rate in the power-law models at early times (see Figure 1). Moreover, a power-law DTD with a long minimum delay may be observationally hard to distinguish from an exponential.

3.1.3. *The Star Formation History*

Figure 5 shows results from one-zone models with a power-law DTD which differ in their assumed minimum SN Ia delay time (t_D) or star formation efficiency timescale (τ_*). In [Fe/H]-[O/Fe] space, doubling or halving the minimum delay time produces similar results to doubling or halving the SFE timescale. A shorter minimum delay time means SNe Ia will start to explode sooner after the star formation burst, giving less time for CCSNe to explode at the plateau of the abundance track. Similarly, a longer SFE timescale means fewer massive stars will be produced by the initial star formation burst, resulting in fewer CCSNe before the first SNe Ia explode.

The minimum delay and SFE timescale have similar effects on the abundance tracks but very different effects on the MDFs. Altering the SFE timescale changes the stellar distribution of [Fe/H], especially at the low-metallicity end, as shown in the top panel of Figure 5.

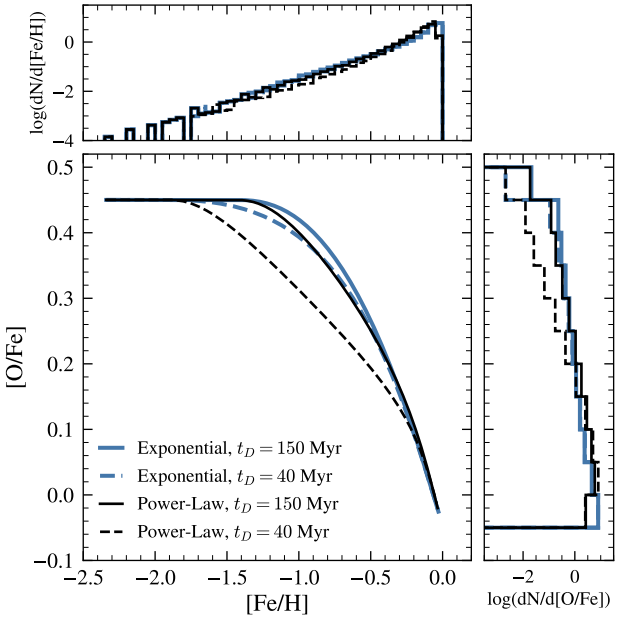

Figure 4. Similar to Figure 2 but with varying minimum SN Ia delay time (t_D). Dashed curves represent models with a minimum delay time of 40 Myr, while solid curves represent models with a minimum delay time of 150 Myr. The black curves represent a power-law DTD with a slope of -1.1, while the blue curves represent an exponential DTD with a timescale of 1.5 Gyr.

However, as a change in the SFE timescale affects both the number of high-mass and low-mass stars produced, it does not affect the distribution of [O/Fe], as shown in the right-hand panel of Figure 5. Meanwhile, a change in the minimum SN Ia delay time does not impact the total number of SNe which explode, so the distribution of [Fe/H] is largely unaffected, while it does result in a change in the number of high-α stars formed and therefore the distribution of [O/Fe].

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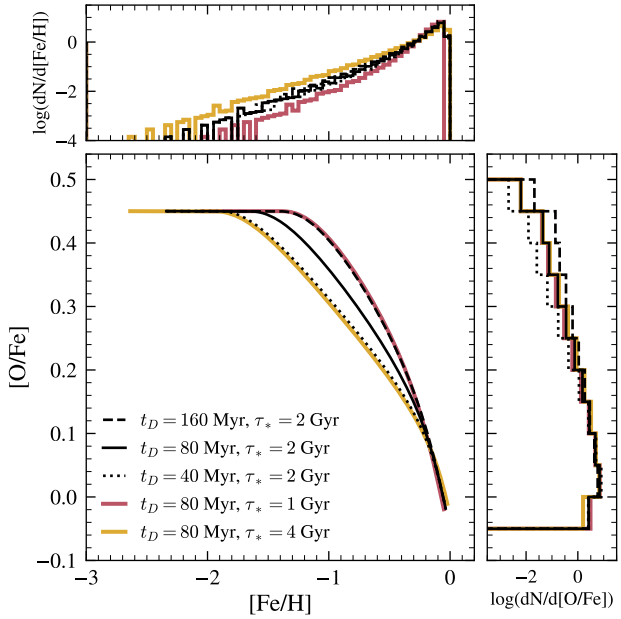

Figure 5. Similar to Figure 2 but with varying minimum SN Ia delay time (t_D) and star formation efficiency timescale (τ_*). Tracks with $\tau_* = 2$ Gyr are represented in black, while a track with a shorter $\tau_* = 1$ Gyr is shown in red and a track with a longer $\tau_* = 4$ Gyr is in yellow. The dotted, solid, and dashed black curves represent a minimum delay time of 40 Myr, 80 Myr, and 160 Myr, respectively. A power-law DTD is assumed in all models.